Let M be a manifold with metric g at TM, $E\to M$ be a vector bundle with metric h, h defines the structure group $O(n)$

(How? vector bundle $E\to M$, at U_α, local frame $\{e_1^\alpha,\cdots,e_n^\alpha\}$, at U_β, $\{e_1^\beta,\cdots,e_n^\beta\}$, at $U_\alpha\cap U_\beta$, $e_i^\beta=(g_{\alpha\beta})_i^j e_j^\alpha$, transition fun $g_{\alpha\beta}(p)$ is a matrix, $g_{\alpha\beta}(p): U_\alpha\cap U_\beta\to GL(n,R)$ is the trans fun, $\mathrm{Im}\, g_{\alpha\beta}$ is G, structure group. Now, E has h, choose an orthnormal frame, $\delta_{ij}=h(e_i^\beta, e_j^\beta)=h((g_{\alpha\beta})_i^k e_k^\alpha,(g_{\alpha\beta})_j^l e_l^\alpha)=(g_{\alpha\beta})_i^k(g_{\alpha\beta})_j^l\delta_{kl}$, since e^α is orthnormal.

$\therefore (g_{\alpha\beta})_i^k(g_{\alpha\beta})_j^k=\delta_{ij}$ $\quad (g_{\alpha\beta})^T(g_{\alpha\beta})=I\to O(n)$)

Let $\{\varphi^i\}$ be an orthnormal frame (for T^*M), $\{\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k}\}_{i_1<\cdots<i_k}$ are orthnormal frame on $\Omega^k(M)$ induced by g ($\langle F,G\rangle=g^{i_1r_1}\cdots g^{i_kr_k}g_{j_1s_1}\cdots g_{j_ls_l}F_{i_1\cdots i_k}^{j_1\cdots j_l}G_{r_1\cdots r_k}^{s_1\cdots s_l}$ for $T_l^k M$)

(~~$\langle\alpha_1\wedge\cdots\wedge\alpha_k,\beta_1\wedge\cdots\wedge\beta_k\rangle_g=\det(\langle\alpha_i,\beta_j\rangle_g)$~~ here, $g_{j_1s_1}\cdots g_{j_ls_l}=1$

$\langle\omega,\eta\rangle=\sum_{a_1\cdots a_k}\sum_{b_1\cdots b_k}g^{a_1b_1}\cdots g^{a_kb_k}\omega_{a_1\cdots a_k}\eta_{b_1\cdots b_k}$ $\quad g^{ab}=\delta^{ab}$, so $\langle\omega,\eta\rangle=\sum_{a_1\cdots a_k}\omega_{a_1\cdots a_k}\eta_{a_1\cdots a_k}$.

$\omega=\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k}$ $\quad \eta=\varphi^{j_1}\wedge\cdots\wedge\varphi^{j_k}$, if $\omega\neq\eta$, $I\neq J$, $\sum\cdots=0+0+1\times0+\cdots+0\times1+0+\cdots=0$

If $\omega=\eta$, $\Sigma=1$)

Let A be the connection compatible with the metric h, curvature tensor $F_A=\sum_{\substack{i<j\\\alpha,\beta}}F_{ij}{}^\alpha{}_\beta\, dx^i\wedge dx^j\otimes e_\alpha\otimes e^\beta$

The point wise norm: $|F_A|^2=\left|\sum_{\substack{i<j\\\alpha,\beta}}F_{ij}{}^\alpha{}_\beta\, dx^i\wedge dx^j\otimes e_\alpha\otimes e^\beta\right|^2=\sum_{\substack{i<j\\k<l\\\alpha,\beta,\gamma,\delta}}F_{ij}{}^\alpha{}_\beta F_{kl}{}^\delta{}_\gamma\, g^{ik}g^{jl}h_{\alpha\delta}h^{\beta\gamma}$

g^{ij}, $h^{\alpha\beta}$ is the inverse matrix of g_{ij}, $h_{\alpha\beta}$

(F_A: 2-form with values in $End(E)$, $F_A=dA+A\wedge A$, $F_A=D_A^2: E=\Omega^0(E)\to\Omega^2(E)$)

Def 1: The Yang-Mills functional

$$YM(A)=\frac{1}{2}\int_M |F_A|^2 dV_g \quad (\text{Volume form } dV_g=\sqrt{\det(g)}\,dx^1\wedge\cdots\wedge dx^n)$$

(Gauge trans: $\underset{\text{vector bundle}}{E\to M}$, the gauge trans is bundle automorphism of E, so it $\in O(n)$, perserve h $h(gs,gt)=h(s,t)$. Now, $A\to g.A$, $F_{g.A}=gF_Ag^{-1}=gF_Ag^T$, g perserve inner product, so $|F_{g.A}|^2=|F_A|^2$ and g only acts on E, not M, so dV_g unchanged, so ~~其~~ YM not change)

$\therefore$ **It is invariant under orthnormal gauge trans.** (so $YM: \underset{\text{moduli space}}{\mathcal{B}_E}=\mathcal{A}_E/\mathcal{G}_E\to R_{\geq0}$, $\mathcal{A}_E$: space of connection, $\mathcal{G}_E$ is the group of gauge trans

Now, we want to find the critical point of this functional (To find geometric structure of $\mathcal{B}_E$) ②
to minize the YM functional.

Riemann-Hilbert correspondence

(For $E \to M$, $\{$flat connections : $F_A = 0\}/$gauge $= \{$reps of $\pi_1(M)\}/$conjugacy. Use holonomy $\leftrightarrow [\gamma]$; $\nabla \to$ parall tran $[\gamma] \to [\gamma]$; represcl

But for some vector bundle, $\exists i$ s.t. $C_i(E) \neq 0$, can not have flat connection. So $|F_A| > 0$, but we need to find $|F_A|_{min}$)
$\left(\frac{1}{2\pi}\int_M F_A\right)$

So, take first variation ~~on connection A~~ of YM

(The first variation of a functional $J(y)$: $\delta J(y,h) = \lim_{t \to 0} \frac{J(y+th) - J(y)}{t} = \frac{d}{dt} J(y+th)\Big|_{t=0}$

Let $A_t = A + ta \quad a \in \Omega^1(\mathfrak{g}_E)$

Let P: principal bundle $\pi: P \to X$, s.t. $P \times G \to P$ perserves fiber. Now G is structure group of P.
$Ad: G \to GL(\mathfrak{g})$. $Ad_g(X) = gXg^{-1}$ $\quad \mathfrak{g}_P := P \times_{Ad} \mathfrak{g} \quad (p, X) \sim (p \cdot g, Ad_{g^{-1}}(X))$
A is a $\mathfrak{g}$ 1-form, $F_A \in \Omega^2(\mathfrak{g}_E)$, Lie algebra $\mathfrak{g}$, very small gauge trans)

Suppose M is cpt (Integral finite, integration by parts works)

$$\frac{d}{dt} YM(A_t)\Big|_{t=0} \overset{cpt}{=} \frac{1}{2}\int_M \frac{d}{dt}\langle F_{A_t}, F_{A_t}\rangle\Big|_{t=0} dV = \int_M \left\langle \frac{d}{dt} F_{A_t}, F_{A_t}\right\rangle\Big|_{t=0} dV$$

$F_A = dA + A\wedge A$. $F_{A_t} = F_A + t\underbrace{(da + A\wedge a + a\wedge A)}_{D_A a} + t^2(a\wedge a)$

$$\boxed{\begin{array}{l} D_A w = dw + A\wedge w + (-1)^{k+1} w\wedge A \\ D_A: \Omega^k(E) \to \Omega^{k+1}(E) \\ \quad \alpha \to d\alpha + A\wedge\alpha \\ D_A: \Omega^k(\mathrm{End}E) \to \Omega^{k+1}(\mathrm{End}E) \end{array}}$$

$$\therefore \ = \int_M \langle D_A a, F_A\rangle dV = \int_M \langle a, \overset{\text{adjoint}}{D_A^*} F_A\rangle dV. \quad \forall a$$

$D_A^*: \Omega^2(\mathfrak{g}_E) \to \Omega^1(\mathfrak{g}_E)$ is formal adjoint of D_A.

Def2: A critical point of it is called YM connection.

i.e. $D_A^* F_A \equiv 0$ (A PDE)

Def3: Let M be a n-dim orientable manifold, let φ^i be an orthonormal frame, the Hodge $*$-operator $*: \Omega^k(M) \to \Omega^{n-k}(M)$ is a linear operator s.t. $*(\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k}) = \pm\varphi^{i_{k+1}}\wedge\cdots\wedge\varphi^{i_n}$
s.t. $(\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k})\wedge(*(\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k})) = dVol$

For this, we have : for real value k-forms α, β, $\langle\alpha,\beta\rangle dV = \alpha\wedge*\beta$

($\alpha = \sum_{i_1<\cdots<i_k} a_{i_1\cdots i_k}\varphi^{i_1}\wedge\cdots\wedge\varphi^{i_k}$, $\beta = \ldots$, $\langle\alpha,\beta\rangle = a_{i_1\cdots i_k} b_{i_1\cdots i_k}$
$\alpha\wedge*\beta = \sum_{I,J} a_I b_J \varphi^I\wedge*\varphi^J = \sum a_I b_J\, dvol = dVol\cdot\langle\alpha,\beta\rangle$.)

If it is a vector bundle E, we define $*: \Omega^k(E) \to \Omega^{n-k}(E)$, acts only on form component, (3)
not the bundle component, to make it linear.

(For $\eta = w^a \otimes s_a$, $w^a = \sum_{1 \le i_1 < \cdots < i_k \le n} w^a_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ (section on $\Lambda^k T^* M$)

$*(w \otimes s) = (*w) \otimes s$)

Pro: $*^2 = (-1)^{k(n-k)}$ on k-forms.

Pf: That is obvious because $\varphi^{i_{k+1}} \wedge \cdots \wedge \varphi^{i_n} \wedge \varphi^{i_1} \wedge \cdots \wedge \varphi^{i_k} = (-1)^{k(n-k)} \varphi^{i_1} \wedge \cdots \wedge \varphi^{i_n}$ ☒

($*(\varphi^{i_1} \wedge \cdots \wedge \varphi^{i_k}) = \varepsilon_1 \varphi^{i_{k+1}} \wedge \cdots \wedge \varphi^{i_n}$ s.t. $\varepsilon_1 \varphi^{i_1} \wedge \cdots \wedge \varphi^{i_n} = dVol = \varphi^1 \wedge \cdots \wedge \varphi^n$. $\varepsilon_1 = \mathrm{sign}(\tau)$

$*^2(\varphi^{i_1} \wedge \cdots \wedge \varphi^{i_k}) = \varepsilon_2 \varepsilon_1 \varphi^{i_1} \wedge \cdots \wedge \varphi^{i_k}$ s.t. $\varepsilon_2 \varphi^{i_{k+1}} \wedge \cdots \wedge \varphi^{i_n} \wedge \varphi^{i_1} \wedge \cdots \wedge \varphi^{i_k} = dVol$

$\varepsilon_1 \varepsilon_2 (-1)^{k(n-k)} \underbrace{\cancel{\varphi^{i_1} \wedge \cdots \wedge \varphi^{i_n}}}_{\varphi^1 \wedge \cdots \wedge \varphi^n} = dVol$ $\therefore \varepsilon_1 \varepsilon_2 = (-1)^{k(n-k)}$)

(We know $\alpha \wedge * \beta = \langle \alpha, \beta \rangle dVol$ for k-form, extend to $\Omega^k(E)$)

Now, consider $\Omega^k(\mathfrak{g}_E)$. Let $E \to M$ be a vector bundle, the structure group is $O(n)$ or its Lie subgroup.
so elements of $\mathfrak{g}$ (Lie algebra $o(n)$: $n \times n$ skew-symmetric) are skew-symmetric matrices. Let $\{e_\alpha\}$ be an orthonormal frame for E.

Review: 0-form
u, v be sections of $\mathfrak{g}_E$, $u = u^\alpha{}_\beta e_\alpha \otimes e^\beta$, $v = v^\alpha{}_\beta e_\alpha \otimes e^\beta$

$\langle u, v \rangle = \left(u^\alpha{}_\beta v^\gamma{}_\delta \delta_{\alpha\gamma} \delta^{\beta\delta} \right) = u^\alpha{}_\beta v^\alpha{}_\beta = -u^\alpha{}_\beta v^\beta{}_\alpha = -\mathrm{Tr}(u \cdot v)$

Then, ~~let~~ $\cancel{w, \eta \in \Omega^k(\mathrm{End} E)}$ $\cong$ $\cancel{E \otimes E^*}$ $\quad w = \underbrace{w^\alpha{}_\beta}_{\text{scalar-valued } k\text{-form}} e_\alpha \otimes e^\beta = \frac{1}{k!} w_{i_1 \cdots i_k}{}^\alpha{}_\beta \, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \otimes e_\alpha \otimes e^\beta$

For $w, \eta \in \Omega^k(\mathfrak{g}_E) \subset \Omega^k(\mathrm{End} E)$. we can use $*$

$\langle w, \eta \rangle dVol = \langle w^\alpha{}_\beta, \eta^\alpha{}_\beta \rangle_{\Omega^k(M)} dVol = w^\alpha{}_\beta \wedge * \eta^\alpha{}_\beta = -w^\alpha{}_\beta \wedge * \eta^\beta{}_\alpha$ $\cancel{= -\mathrm{Tr}\, w \wedge * \eta}$

(wedge ~~inner~~ product of endomorphism-valued forms, $w \wedge \eta = (w^\alpha{}_\gamma \wedge \eta^\gamma{}_\beta) e_\alpha \otimes e^\beta$), $\mathrm{Tr}\, w \wedge * \eta$

$= (w \wedge (*\eta))^\alpha_\alpha = w^\alpha{}_\beta \wedge * \eta^\beta{}_\alpha$) ($(*\eta)^\beta_\alpha = *(\eta^\beta_\alpha)$ because $*$ only act on form component, instead of bundle component)

$\therefore \langle w, \eta \rangle dVol = -\mathrm{Tr}\, w \wedge * \eta$

(So we get the formula for Lie-algebra valued form)

Then, we can get:

Lemma 1. The formal adjoint $D_A^*: \Omega^k(g_E) \to \Omega^{k-1}(g_E)$ is given by

$$D_A^* = (-1)^{1+n(k-1)} * D_A *$$

and
$$\int_M \langle D_A\mu, \omega\rangle \, dV = \int_M \langle \mu, D_A^*\omega\rangle \, dV.$$

for compactly supported forms $\mu \in \Omega^{k-1}(g_E)$ and $\omega \in \Omega^k(g_E)$

Pf: $d\, Tr(\mu \wedge *\omega) = Tr(D_A\mu \wedge *\omega) + (-1)^{k-1} Tr(\mu \wedge D_A *\omega)$

Integral

$$\int d\, Tr(\mu\wedge *\omega) \overset{\text{stokes}}{=} 0 = \int Tr(D_A\mu \wedge *\omega) + (-1)^{k-1} Tr(\mu \wedge D_A *\omega)$$

From above $Tr(D_A\mu \wedge *\omega) = -\langle D_A\mu, \omega\rangle dV$

$$\therefore \int \langle D_A\mu, \omega\rangle dV = (-1)^{k-1} \int Tr(\mu \wedge D_A *\omega)$$

$$= (-1)^k \int \langle \mu, *^{-1} D_A *\omega\rangle dV$$

($*^2 = (-1)^{k(n-k)}$ for k-form, $D_A*\omega$ is $n-k+1$ form)

$$= (-1)^{k+(n-k+1)(k-1)} \int \langle \mu, * D_A * \mu\rangle dV.$$

$\because k+(n-k+1)(k-1) \equiv (n-k+2)(k-1)+1 \equiv (n-k)(k-1)+1 \equiv nk-k^2+k-n+1 \mod 2.$

$k^2-k = k(k-1) \equiv 0 \mod 2$, $nk-n+1 \equiv nk+n+1 \mod 2$

$$\therefore \int \langle D_A\mu, \omega\rangle dV = \int (-1)^{n(k+1)+1} \langle \mu, * D_A * \omega\rangle dV$$

$$\therefore D_A^* = (-1)^{n(k+1)+1} * D_A *$$

∎

Recall that YM connection $D_A^* F_A = 0$, means $\pm * D_A * F_A = 0 \Rightarrow D_A * F_A = 0$

($\pm *$ is iso, linear, inj, sur)

So we get
$$\begin{cases} D_A F_A = 0 & \text{(Bianchi identity)} \\ D_A * F_A = 0 \end{cases}$$

Another expression for the formal adjoint. [redacted] (For future calculations)

$$D_A : \Omega^k(g_E) \xrightarrow{\nabla_A} T^*M \otimes \Omega^k(g_E) \xrightarrow{\wedge} \Omega^{k-1}(g_E)$$

Let E be a vector bundle, metric h, connection A. covariant derivitive ∇_A (for tensor field)

Def $\Omega^k(E) = \Gamma(\Lambda^k T^*M \otimes E)$ be k-form space, $\Omega^k(E) \subset \Gamma(T^*M^{\otimes k} \otimes E)$

Let a k-form $\omega = \sum_{i_1 < \cdots < i_k} \omega_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \frac{1}{k!} \omega_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ ($\omega_{i_1 \cdots i_k}$ is alternating)
(all values of $i_1, \ldots, i_k$)

$v = v^i e_i \to \exp_p(v)$

Choose geodesic coordinates at $p \in M$, s.t. at p, $g_{ij} = \delta_{ij}$, Christoffel symbol $= 0$.

$$\langle \omega, \eta \rangle = \sum_{i_1 < \cdots < i_k} \langle \omega_{i_1 \cdots i_k}, \eta_{i_1 \cdots i_k} \rangle_g = \frac{1}{k!} \langle \omega_{i_1 \cdots i_k}, \eta_{i_1 \cdots i_k} \rangle_g$$

Let $k-1$-form $\mu = \frac{1}{(k-1)!} \mu_{i_2 \cdots i_k} dx^{i_2} \wedge \cdots \wedge dx^{i_k}$, $D_A \mu$, a k-form, at geodesic coordinates s.t. $D_A \mu = \frac{1}{(k-1)!} \nabla_{i_1} \mu_{i_2 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ ($D_A = \wedge \circ \nabla_A$, $\nabla_A \mu = dx^i \otimes \nabla_i \mu$, $D_A \mu = dx^i \wedge \nabla_i \mu = \nabla_{i_1} \mu_{i_2 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$)
$\mu = \mu_{j_1 \cdots j_{k-1}}$
$i \to i_1$

However, we need the coefficience alternating.

so $D_A \mu = \frac{1}{k!} (\nabla_{i_1} \mu_{i_2 \cdots i_k} - \nabla_{i_2} \mu_{i_1 i_3 \cdots i_k} - \cdots - \nabla_{i_k} \mu_{i_2 \cdots i_{k-1} i_1}) dx^{i_1} \wedge \cdots \wedge dx^{i_k}$

$(D_A \mu)_{i_1, \ldots, i_k} = \nabla_{[i_1} \mu_{i_2 \cdots i_k]} = \frac{1}{k} \sum_{j=1}^{k} (-1)^{j-1} \nabla_{i_j} \mu_{i_1 \cdots \hat{i_j} \cdots i_k}$. , $\mu_{i_1 \cdots i_k}$ alternating.

$\Uparrow$

$k \nabla_{i_1} \mu_{i_2 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \nabla_{i_1} \mu_{i_2 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} + \underline{\nabla_{i_2} \mu_{i_1 i_3 \cdots i_k} dx^{i_2} \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}} + \cdots$
$+ \nabla_{i_k} \mu_{i_2 \cdots i_{k-1} i_1} dx^{i_k} \wedge \cdots \wedge dx^{i_1}$

用 $(i_2 i_1 i_3 \cdots i_k)$ 词条代替 $(i_1 i_2 \cdots i_k)$.

$i_1 \leftrightarrow i_j$

$= \nabla_{i_1} \mu_{i_2 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} - \nabla_{i_2} \mu_{i_1 i_3 \cdots i_k} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k} - \cdots$

$$\langle D_A \mu, \omega \rangle = \frac{1}{k!} \langle \nabla_{i_1} \mu_{i_2 \cdots i_k} - \nabla_{i_2} \mu_{i_1 i_3 \cdots i_k} - \cdots - \nabla_{i_k} \mu_{i_2 \cdots i_1}, \omega_{i_1 \cdots i_k} \rangle$$

$= \frac{1}{(k-1)!} \langle \nabla_{i_1} \mu_{i_2 \cdots i_k}, \omega_{i_1 \cdots i_k} \rangle$ (change $i_{j_1} i_{j_2}$ in $\omega_{i_1 \cdots i_k}$; $\omega_{i_1 \cdots i_k}$ is alternating)

On the other hand.

$\langle D_A \mu, \omega \rangle = \underbrace{\frac{1}{(k-1)!} (\nabla_{i_1} \langle \mu_{i_2 \cdots i_k}, \omega_{i_1 i_2 \cdots i_k} \rangle}_{\text{div term}} - \langle \mu_{i_2 \cdots i_k}, \nabla_{i_1} \omega_{i_1 \cdots i_k} \rangle)$ (Ricci lemma, L-Cconnection $\nabla_i g_{jk} = 0$)

$\Downarrow$ div term; By stokes, integral $= 0$

$\Downarrow$ scalar-fun.

$= \text{div} \sim - \frac{1}{(k-1)!} \langle \mu_{i_2 \cdots i_k}, \nabla_{i_1} \omega_{i_1 \cdots i_k} \rangle$

and $\because \langle \mu, D_A^* \omega \rangle = \frac{1}{(k-1)!} \langle \mu_{i_2 \cdots i_k}, (D_A^* \omega)_{i_2 \cdots i_k} \rangle$ (μ is $k-1$ form)

$\therefore (D_A^* \omega)_{i_2 \cdots i_k} = -\nabla_{i_1} \omega_{i_1 \cdots i_k} = -\nabla_i \omega_{i\, i_2 \cdots i_k} = -g^{ij} \nabla_i \omega_{j\, i_2 \cdots i_k}$.

YM eq: $D_A^* F_A = 0$ $\quad \therefore \nabla^i F_{ik} = 0$ (F_A is 2-form)

($F_{ij} = -F_{ji}$, $\nabla^j = g^{ij} \nabla_i$)

Maxwell's Equations

Ex1: In R^3

Let $G = U(1)$ (Abel group), Lie algebra $g = iR$

$U(1): \{z \in \mathbb{C}, |z| = 1\} \Rightarrow z = e^{i\theta}$ $\quad \gamma(t) = e^{i\phi(t)}$, $\phi(0) = 0$, $\left.\frac{d\gamma}{dt}\right|_{t=0} = i\phi'(0) \to iR$ (切空间)

$[X, Y] = 0$.

Connection $A = i(A_1 dx^1 + A_2 dx^2 + A_3 dx^3)$ $\quad \vec{A} = (A_1, A_2, A_3)$ is called vector potential.

Curvature $F = dA = i[(\partial_1 A_2 - \partial_2 A_1) dx^1 \wedge dx^2 + (\partial_1 A_3 - \partial_3 A_1) dx^1 \wedge dx^3 + (\partial_2 A_3 - \partial_3 A_2) dx^2 \wedge dx^3]$ (G is Abel)

($F_A = dA + A \wedge A$, $A \wedge \omega$: $A = A_\mu dx^\mu$, $\omega = \frac{1}{k!} \omega_{\nu_1 \cdots \nu_k} dx^{\nu_1} \wedge \cdots \wedge dx^{\nu_k}$, $[A \wedge \omega] = \frac{1}{k!} [A_\mu, \omega_{\nu_1 \cdots \nu_k}] dx^\mu \wedge dx^{\nu_1} \wedge \cdots \wedge dx^{\nu_k}$)

As Lie algebra form

Let magnetic field $\vec{B} = (B_1, B_2, B_3)$, defined by:

$$F_{ij} = \sqrt{-1} \begin{pmatrix} 0 & B_3 & -B_2 \\ -B_3 & 0 & B_1 \\ B_2 & -B_1 & 0 \end{pmatrix}$$

$B_1 = \partial_2 A_3 - \partial_3 A_2$, $B_2 = \partial_3 A_1 - \partial_1 A_3$, $B_3 = \partial_1 A_2 - \partial_2 A_1$ (都是实数)

($F = \frac{1}{2} F_{ij} dx^i \wedge dx^j$)

$\therefore \vec{B} = \nabla \times \vec{A}$, $F = i(B_1 dx^2 \wedge dx^3 + B_2 dx^1 \wedge dx^3 + B_3 dx^1 \wedge dx^2)$

$*F = i(B_1 dx^1 + B_2 dx^2 + B_3 dx^3)$

From Y-M equation: $d * F = 0$ $\quad$ ($U(1)$ Abel, $D_A \omega = d\omega + \underbrace{[A \wedge \omega]}_{0}$, $\therefore D_A = d$)

$\therefore i[(\partial_1 B_2 - \partial_2 B_1) dx^1 \wedge dx^2 + (\partial_1 B_3 - \partial_3 B_1) dx^1 \wedge dx^3 + \cancel{(\partial_3 B_2 - \partial_2 B_3)\, d} (\partial_2 B_3 - \partial_3 B_2) dx^2 \wedge dx^3] = 0$

$\Leftrightarrow \nabla \times \vec{B} = 0$

From Bianchi equation: $dF = 0$.

$i(\partial_1 B_1 dx^1 \wedge dx^2 \wedge dx^3 + \partial_2 B_2 dx^2 \wedge dx^1 \wedge dx^3 + \partial_3 B_3 dx^3 \wedge dx^1 \wedge dx^2) = 0$

$\Leftrightarrow i(\partial_1 B_1 + \partial_2 B_2 + \partial_3 B_3) dx^1 \wedge dx^2 \wedge dx^3 = 0 \Leftrightarrow \underbrace{\nabla \cdot \vec{B}}_{(\mathrm{div}\, B)} = 0$.

$\therefore \begin{cases} \nabla \times \vec{B} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases}$ Maxwell's eq with $\vec{E} = 0$ (electric field) (in vacuum)

(gauge invariance: $\vec{B}$ is unchanged when $A \to A + du$)

Ex2. Minkowski space $R^{3,1}$ $A=i(\rho dt+A_1dx^1+A_2dx^2+A_3dx^3)$ ρ is electric potential

$$F=dA=i(d\rho\wedge dt+dA_1\wedge dx^1+dA_2\wedge dx^2+dA_3\wedge dx^3)$$

$$=i\left(\frac{\partial\rho}{\partial x^1}dx^1\wedge dt+\frac{\partial\rho}{\partial x^2}dx^2\wedge dt+\frac{\partial\rho}{\partial x^3}dx^3\wedge dt+\frac{\partial A_1}{\partial x_2}dx^2\wedge dx^1+\frac{\partial A_1}{\partial t}dt\wedge dx^1+\frac{\partial A_1}{\partial x^3}dx^3\wedge dx^1\right.$$
$$\left.+\frac{\partial A_2}{\partial t}dt\wedge dx^2+\frac{\partial A_2}{\partial x^1}dx^1\wedge dx^2+\frac{\partial A_2}{\partial x^3}dx^3\wedge dx^2+\frac{\partial A_3}{\partial t}dt\wedge dx^3+\frac{\partial A_3}{\partial x^1}dx^1\wedge dx^3+\frac{\partial A_3}{\partial x^2}dx^2\wedge dx^3\right)$$

$$=i\left[\left(\frac{\partial A_1}{\partial t}-\frac{\partial\rho}{\partial x^1}\right)dt\wedge dx^1+\cdots+\left(\frac{\partial A_2}{\partial x^1}-\frac{\partial A_1}{\partial x^2}\right)dx^1\wedge dx^2+\cdots\right]$$

$$=i\left[\left(\frac{\partial A_j}{\partial t}-\frac{\partial\rho}{\partial x^j}\right)dt\wedge dx^j+\frac{1}{2}\left(\frac{\partial A_j}{\partial x^k}-\frac{\partial A_k}{\partial x^j}\right)dx^k\wedge dx^j\right]$$

Def $\vec{E},\vec{B}$ by

$$F_{ij}=\sqrt{-1}\begin{pmatrix}0&-E_1&-E_2&-E_3\\E_1&0&B_3&-B_2\\E_2&-B_3&0&B_1\\E_3&B_2&-B_1&0\end{pmatrix}$$

$E_j:=\frac{\partial\rho}{\partial t}-\frac{\partial A_j}{\partial t}$ $\Rightarrow \vec{E}=\nabla\rho$ (If A is independent to t)

$B_1:=\frac{\partial A_3}{\partial x^2}-\frac{\partial A_2}{\partial x^3}$, $B_2=\frac{\partial A_1}{\partial x^3}-\frac{\partial A_3}{\partial x^1}$, $B_3=\frac{\partial A_2}{\partial x^1}-\frac{\partial A_1}{\partial x^2}$

$\Rightarrow \vec{B}=\nabla\times\vec{A}$

Bianchi eq: $dF=0$ $\left(d\left(\frac{1}{2}F_{ij}dx^i\wedge dx^j\right)=\frac{1}{2}\left(\partial_kF_{ij}dx^k\wedge dx^i\wedge dx^j+\partial_jF_{ki}dx^j\wedge dx^k\wedge dx^i+\partial_iF_{jk}dx^i\wedge dx^j\wedge dx^k\right)=0\right)$

$\Rightarrow \partial_iF_{jk}+\partial_jF_{ki}+\partial_kF_{ij}=0$

Let $(i,j,k)=(1,2,3)$, $\frac{\partial F_{23}}{\partial x^1}+\frac{\partial F_{31}}{\partial x^2}+\frac{\partial F_{12}}{\partial x^3}=\frac{\partial B_1}{\partial x^1}+\frac{\partial B_2}{\partial x^2}+\frac{\partial B_3}{\partial x^3}=\nabla\cdot\vec{B}=0$ (div $\vec{B}$) (Gauss's Law of magnetism)

Let $(i,j,k)=(0,1,2)$, $\frac{\partial B_3}{\partial t}+\frac{\partial E_2}{\partial x^1}-\frac{\partial E_1}{\partial x^2}=0$ $\quad(\nabla\times\vec{E})_3=-\frac{\partial B_3}{\partial t}$

Let $(i,j,k)=(0,2,3)$, $\frac{\partial B_1}{\partial t}+\frac{\partial E_3}{\partial x^2}-\frac{\partial E_2}{\partial x^3}=0$ $\quad(\nabla\times\vec{E})_1=-\frac{\partial B_1}{\partial t}$ $\Rightarrow \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$ (Faraday's Law)

Let $(i,j,k)=(0,3,1)$, $\frac{\partial B_2}{\partial t}+\frac{\partial E_1}{\partial x^3}-\frac{\partial E_3}{\partial x^1}=0$ $\quad(\nabla\times\vec{E})_2=-\frac{\partial B_2}{\partial t}$

YM-eq: $d*F=0$, Choose west coast Minkowski signature $(+,-,-,-)$

$\nabla^iF_{ik}=0$, i.e. $g^{ij}\nabla_iF_{jk}=0$ $\quad g^{ij}=\begin{pmatrix}1&&&\\&-1&&\\&&-1&\\&&&-1\end{pmatrix}$

$(\because g^{00}\nabla_0F_{0k}+g^{11}\nabla_1F_{1k}+g^{22}\nabla_2F_{2k}+g^{33}\nabla_3F_{3k}=0)$

$k=0$: $1\times\frac{\partial F_{00}}{\partial t}-\frac{\partial F_{10}}{\partial x^1}-\frac{\partial F_{20}}{\partial x^2}-\frac{\partial F_{30}}{\partial x^3}=0$, $-\frac{\partial E_1}{\partial x^1}-\frac{\partial E_2}{\partial x^2}-\frac{\partial E_3}{\partial x^3}=0$ $\quad\nabla\cdot\vec{E}=0$ (Gauss's Law)

$k=1$: $\frac{\partial F_{01}}{\partial t}-\frac{\partial F_{11}}{\partial x^1}-\frac{\partial F_{21}}{\partial x^2}-\frac{\partial F_{31}}{\partial x^3}=0$, $-\frac{\partial E_1}{\partial t}+\frac{\partial B_3}{\partial x^2}-\frac{\partial B_2}{\partial x^3}=0$, $\frac{\partial E_1}{\partial t}=(\nabla\times\vec{B})_1$

$k=2$: $-\frac{\partial E_2}{\partial t}-\frac{\partial B_3}{\partial x^1}+\frac{\partial B_1}{\partial x^3}=0$, $\frac{\partial E_2}{\partial t}=(\nabla\times\vec{B})_2$

$k=3$: $-\frac{\partial E_3}{\partial t}+\frac{\partial B_2}{\partial x^1}-\frac{\partial B_1}{\partial x^2}=0$, $\frac{\partial E_3}{\partial t}=(\nabla\times\vec{B})_3$

$\therefore \nabla\times\vec{B}=\frac{\partial\vec{E}}{\partial t}$ (Ampère's Law)

$dF=0\Rightarrow \begin{cases}\nabla\cdot\vec{B}=0\\ \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}\end{cases}$ $\quad d*F=0\Rightarrow\begin{cases}\nabla\cdot\vec{E}=0\\ \nabla\times\vec{B}=\frac{\partial\vec{E}}{\partial t}\end{cases}$ Maxwell eqs.

Dirac equation:

On $\mathbb{R}^{3,1}$, ψ is a spinor field (4-complex vector). A is a connection ($A = i(\phi dt + A_1 dx^1 + A_2 dx^2 + A_3 dx^3)$)

Dirac uses ψ describe a wave function of an electron in electromagnetic field A, with equation: $i \not{\partial}_A \psi = m\psi$ $\quad$ m: mass $\quad \not{\partial}_A = \gamma^\mu \cdot (\partial_\mu + A_\mu)$. γ^μ is dirac matrix, satisfies Clifford algebra, $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \cdot I$ $\quad (\eta^{\mu\nu} = (1,-1,-1,-1))$

Dirac thought A should have physical meaning since it is a part of equation.

Ex3. (magnetic monopole) (physical model)

In $\mathbb{R}^3 \setminus \{0\}$, $\vec{E} \equiv 0$, $\vec{B} = \frac{q\vec{r}}{r^3} = q \cdot \frac{x^i dx^i}{r^3}$

$$F = i[\underbrace{B_3 dx^1 \wedge dx^2 - B_2 dx^1 \wedge dx^3 + B_1 dx^2 \wedge dx^3}_{*B}] \qquad *F = i\begin{pmatrix} 0 & B_1 & B_2 & B_3 \\ -B_1 & & & \\ -B_2 & & 0 & \\ -B_3 & & & \end{pmatrix}$$

$-i*F = \vec{B} = q\frac{x^i dx^i}{r^3} = -q\, d(\frac{1}{r})$, closed. $\quad \therefore d*F = 0$.

$-i dF = d*B = -q\, d*d(\frac{1}{r}) = -q * \Delta(\frac{1}{r}) = 0$ $\quad$ ($\frac{1}{r}$: harmonic on $r>0$).

$(\Delta = *d*d,\ *\Delta = *^2 d*d,\ *^2 = (-1)^{3\times(3-3)} = 1)$

$$\left(\begin{array}{l} d: f_1 dx^1 + \cdots + f_n dx^n;\ *d: f_1 dx^2 \wedge \cdots \wedge dx^n - f_2 dx^1 \wedge dx^3 \wedge \cdots \wedge dx^n + (-1)^{n-1} f_n dx^1 \wedge \cdots \wedge dx^{n-1}; \\ d*d: f_{11} dx^1 \wedge \cdots \wedge dx^n - f_{22} dx^2 \wedge dx^1 \wedge dx^3 \wedge \cdots \wedge dx^n + (-1)^{n-1} f_{nn} dx^n \wedge dx^1 \wedge \cdots \wedge dx^{n-1} = (f_{11} + \cdots + f_{nn}) dx^1 \wedge \cdots \wedge dx^n \\ *d*d = f_{11} + \cdots + f_{nn} \end{array}\right)$$

$\therefore dF = 0$ $\quad \therefore$ This is a soluation of Maxwell's equations.

So, Dirac had a question: Does every soluation have physical meaning? We have $\vec{B}, \vec{E}$, so $\vec{F}$, but is there always a connection (vector potiential) s.t. F is the curvature of A?

$F = i*B = \frac{iq}{r^3} * x^i dx^i = iq \sin\theta\, d\theta \wedge d\varphi$ (sphere coordinate, $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$)

Take charts $U_0 = \mathbb{R}^3 - \{(0,0,z), z \le 0\}$ (negative z-axis), $U_1 = \mathbb{R}^3 - \{(0,0,z), z \ge 0\}$ (positive z-axis)

On U_0, $A_0 = iq(1-\cos\theta)d\varphi$ (So it can extend smoothly to positive z-axis)

On U_1, $A_1 = -iq(1+\cos\theta)d\varphi$ (extend smoothly to negative z-axis) $\quad \Rightarrow dA_0 = F$, $dA_1 = F$

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But in real world, observe a same thing in two places, its physical condition should be same, i.e. the wavefunction should be same.

But for A_0, A_1, they satisfy different Dirac equations, $i\not{D}_{A_0}\psi_0 = m\psi_0$ in U_0; $i\not{D}_{A_1}\psi_1 = m\psi_1$ in U_1. How to let them describe the same electron? $\psi_1(x) = G(x)\psi_0(x)$. $G(x)$ is $U(1)$-valued.

So: $\cancel{\not{D}_\mu \psi_1 = G\not{D}_\mu \psi_0,\ \psi_0 = G\psi_1}$ $\quad \not{D}_A = \gamma^\mu(\partial_\mu + A_\mu)$

$$[(\partial_\mu G)\psi_0 + G\partial_\mu\psi_0 + A_{1\mu}G\psi_0]\, i\gamma^\mu = m\, G\psi_0$$ ↙ Dirac eq on U_1, $\psi_1 = G\psi_0$,

$$i\gamma^\mu G[\partial_\mu\psi_0 + (G^{-1}\partial_\mu G + A_{1\mu})\psi_0] = mG\psi_0 = G\cdot i\gamma^\mu(\partial_\mu\psi_0 + A_{0\mu}\psi_0)$$ ↙ Dirac eq on U_0

$$\therefore\ \partial_\mu\psi_0 + (G^{-1}\partial_\mu G + A_{1\mu})\psi_0 = \partial_\mu\psi_0 + A_{0\mu}\psi_0$$

$$\therefore\ G^{-1}\partial_\mu G + A_{1\mu} = A_{0\mu} \qquad A_0 - A_1 = G^{-1}dG = d\log G.$$

We know: $A_0 - A_1 = 2iq\,d\varphi = d(2iq\varphi)$ $\quad\therefore\ G = c\cdot e^{2iq\varphi}$.

To make G well-defined, ($\psi_1 = G\psi_0$, ψ_1, ψ_0 are wave-functions, G has period)

$$c\cdot e^{2iq(\varphi+2\pi)} = c\cdot e^{2iq(\varphi)} \quad \therefore\ e^{4\pi iq} = 1 \qquad 2q \in \mathbb{Z}$$

$$\left(c_1 = \frac{i}{2\pi}\int_{S^2} F = \frac{i}{2\pi}\int iq\sin\theta\, d\theta\wedge d\varphi = 4\pi i q\frac{i}{2\pi} = -2q \in \mathbb{Z}\right).$$

$c_1 = [\frac{i}{2\pi}\mathrm{Tr}_{\mathbb{C}} F_A]$

$\therefore$ The anwser is no, there is another condition: $2q \in \mathbb{Z}$, it's called Quantization of Electric charge, i.e. the charge of electronic body must be an integer multiple of a certain charge - elementary charge $e = 1.602\times10^{-19}\,C$ (Millikan Oil-drop experiment, 1909)